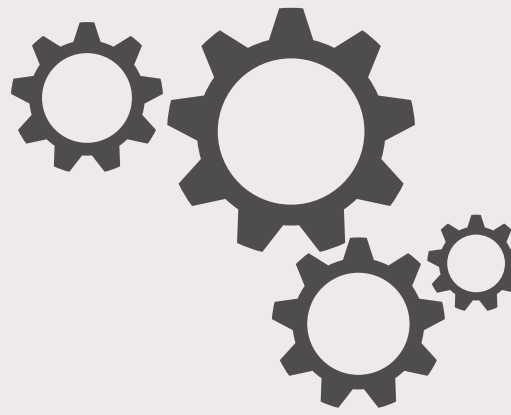
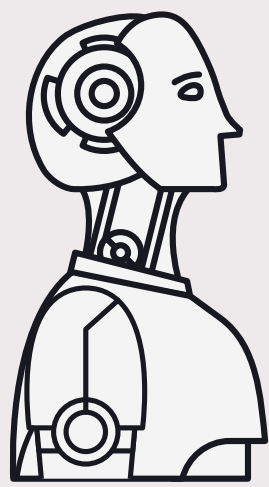
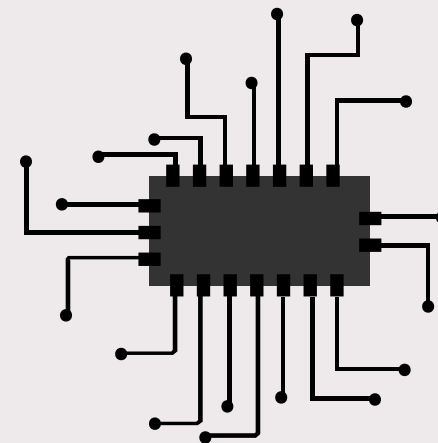




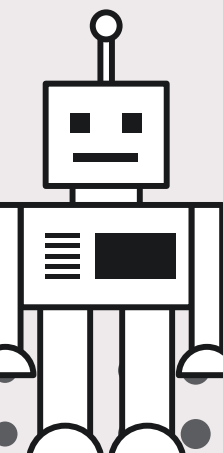
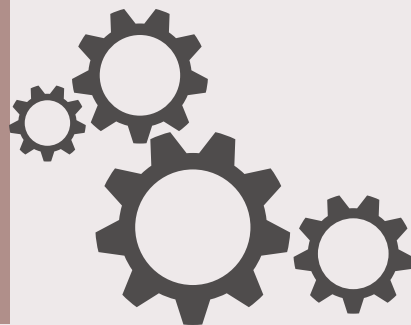
LINE FOLLOWER ROBOT

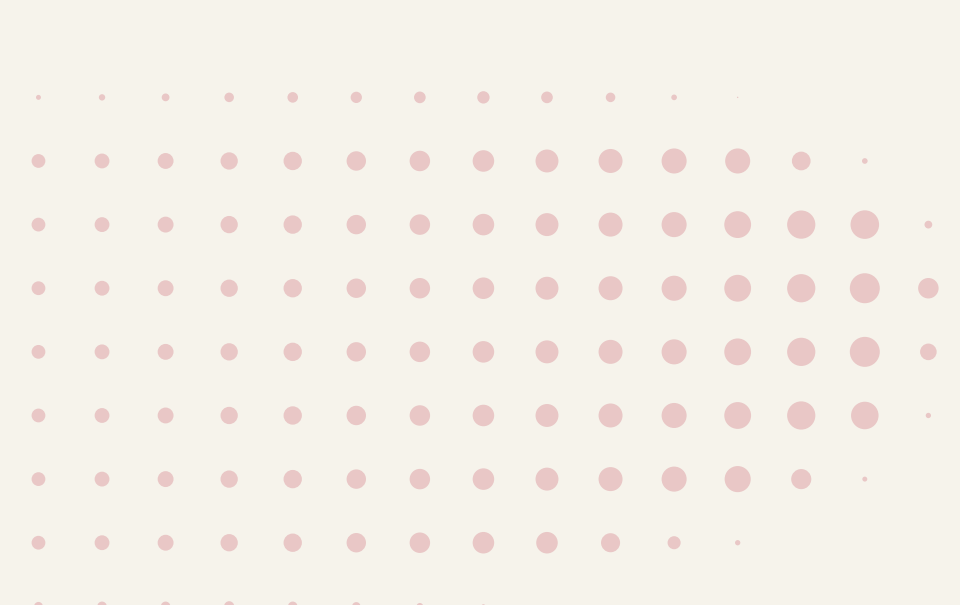
Batch: B3



Group: E

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OUR TEAM

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ROLL NUMBER

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WORKING

The concept of the line follower robot is related to light. Here, we use the behavior of light on the black-and-white surface. The white color reflects all the light that falls on it, whereas the black color absorbs the light.

In this line-follower robot, we use IR transmitters and receivers (photodiodes). They are used to send and receive the lights. When IR rays fall on a white surface, they are reflected toward the IR receiver, generating some voltage changes.

WORKING

Black surfaces absorb infrared radiation and do not reflect any of the rays that fall on them; therefore, no photons reach the infrared receiver.

In this project, when the IR sensor senses a white surface, an Arduino gets 1 (high) as input, and when it senses a black line, an Arduino gets 0 (low) as input. Based on these inputs, an Arduino Uno provides the proper output to control the bot.

WORKING

Our group's line follower project goes beyond conventional functionality by incorporating a seed dropping mechanism. While autonomously tracing lines, it also disperses seeds along its path, offering an innovative solution for agricultural tasks.

The seed dropper mechanism operates efficiently through a cascading series of mechanical actions: as the line follower wheel rotates, it engages with another mechanical component, aligning perforations to enable seed release at precise intervals.

COMPONENTS USED IN LINE FOLLOWER ROBOT

5

Motor System

BO-motors - Enables the robot to move forward, backward, and turn

Sensors

The infrared sensor accurately detects heat radiation, enabling precise measurements

Microcontroller

Arduino - Acts as the brain of the robot, processing sensor data and controlling motor actions

Batteries

Rechargeable Li-ion cell AJT
18650-20 2000MAH 3.7V * 2 -
Powers the robot, providing long-lasting performance

PROCEDURE

1**Wheels of the line follower:**

Design a 3D model of a wheel using the Autodesk Fusion 360 software .
Keep in mind the dimensions of the motors to design the wheels.
Print it out using the 3D printer.

2**Laser cutting the chassis:**

Sketch the design of the chassis that you want in the Autodesk Fusion 360.
Cut the acrylic sheet in that design using the laser cutter.

3**Making Circuit Connections:**

make the circuit connections joining all of the electrical components of the line follower robot according the image on the next slide.

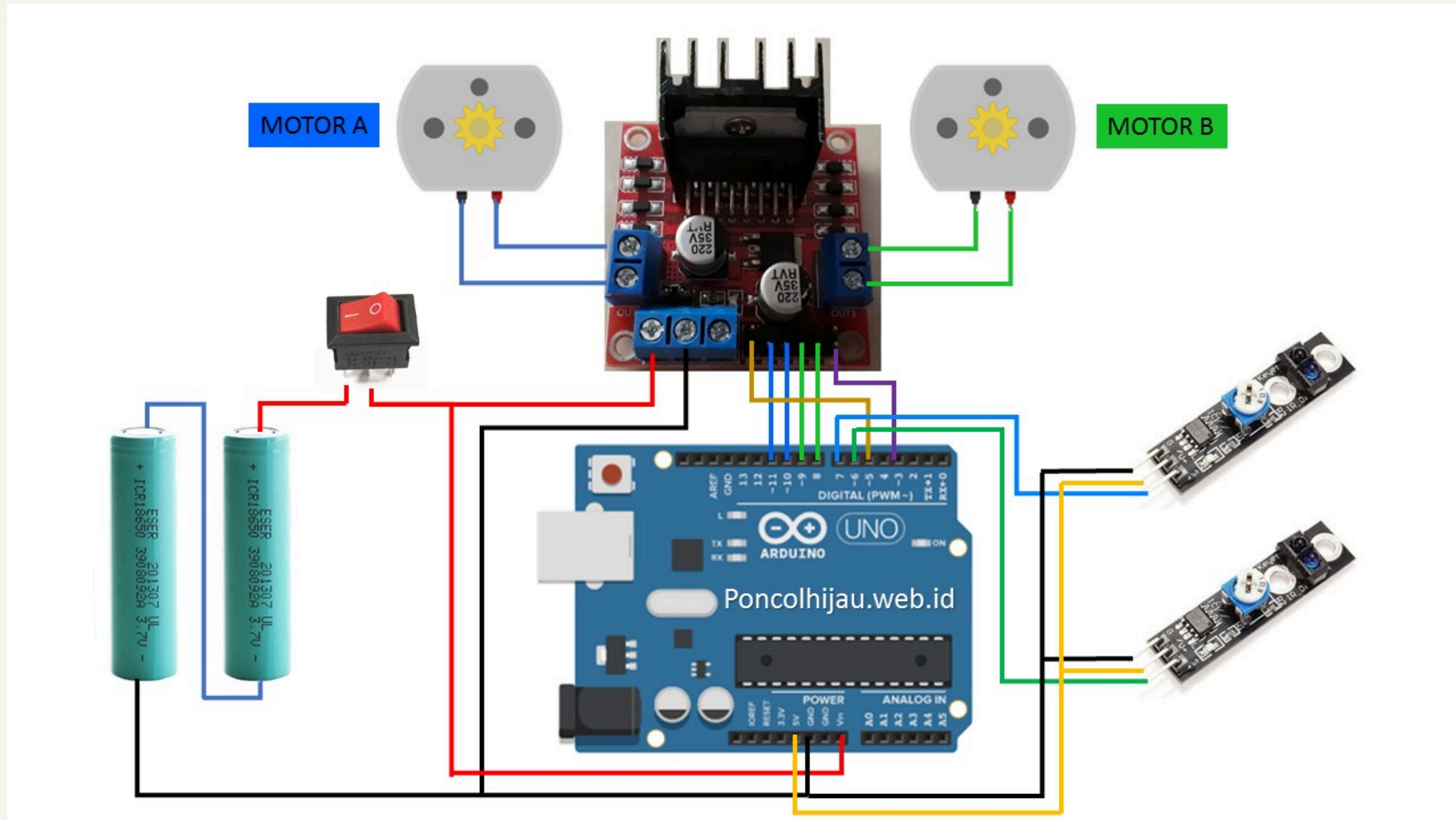


Fig: The Circuit Connections

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PROCEDURE

3

Coding for the Microcontroller:

The code was developed with Arduino IDE by following the logic on which the line follower is based on. Connect the microcontroller to your computer using the programming cable. Upload the written code to your microcontroller.

4

Assembling the components of the chassis:

Make rough sketch of how you will assemble all the components. Now attach the motors on the chassis using double sided tape. Attach the wheels to the motors.

5

Connecting the IR sensors to Arduino Uno.:

IR sensors should be attached in such a way such that the LEDs face downwards. The gap between the IR sensors should not be too large or too small. This distance can be ascertained by trial and error to see which distance allows the robot to follow the path most accurately.

PROCEDURE

6**Testing:**

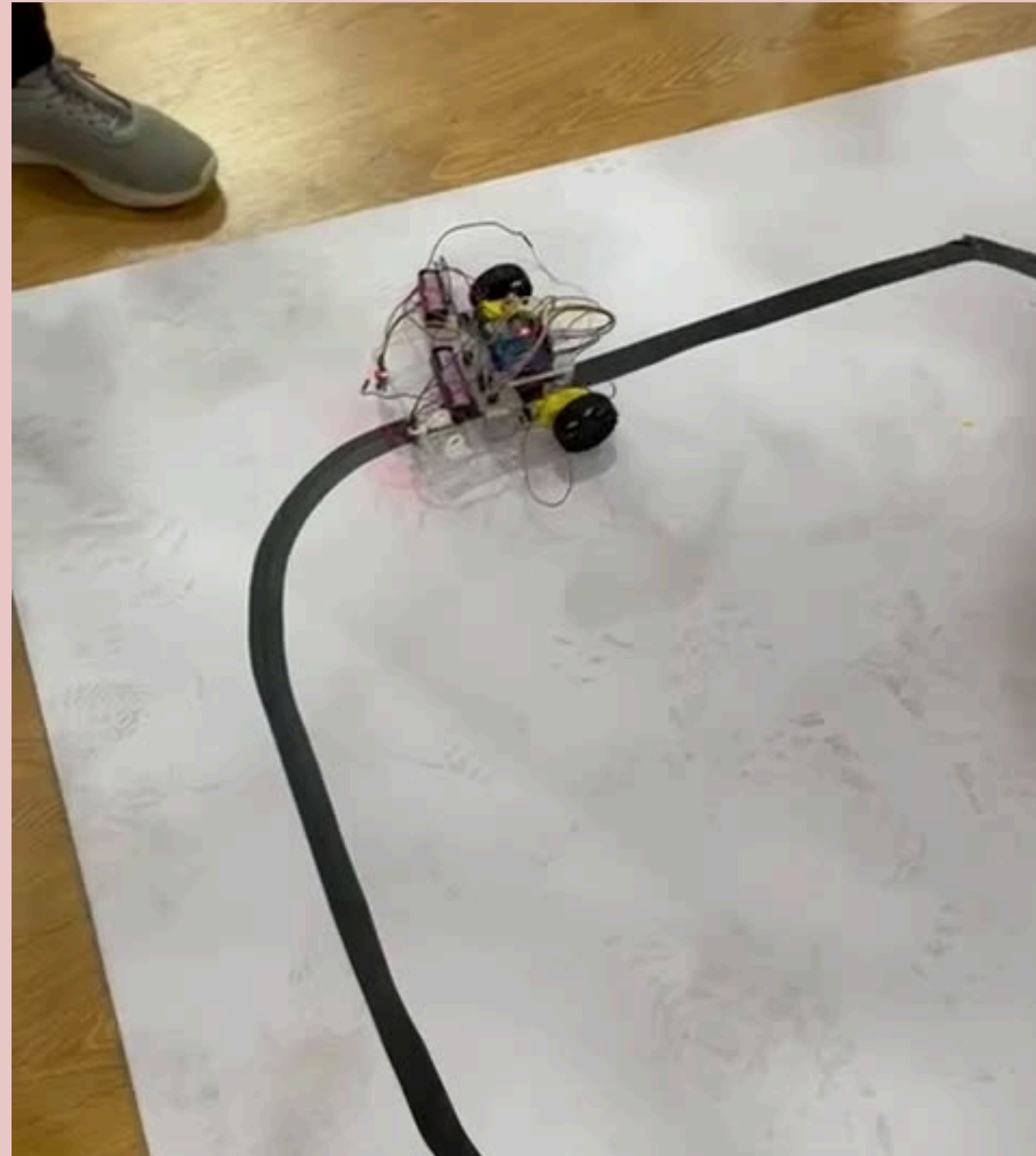
Test your finished line follower on a black line drawn on a white surface.

During the testing phase, we encountered challenges and fine tuned the code to optimize the robot's performance and address any issues.

After rigorous debugging, we achieved a robust codebase that enables the line follower robot to track and follow lines accurately.

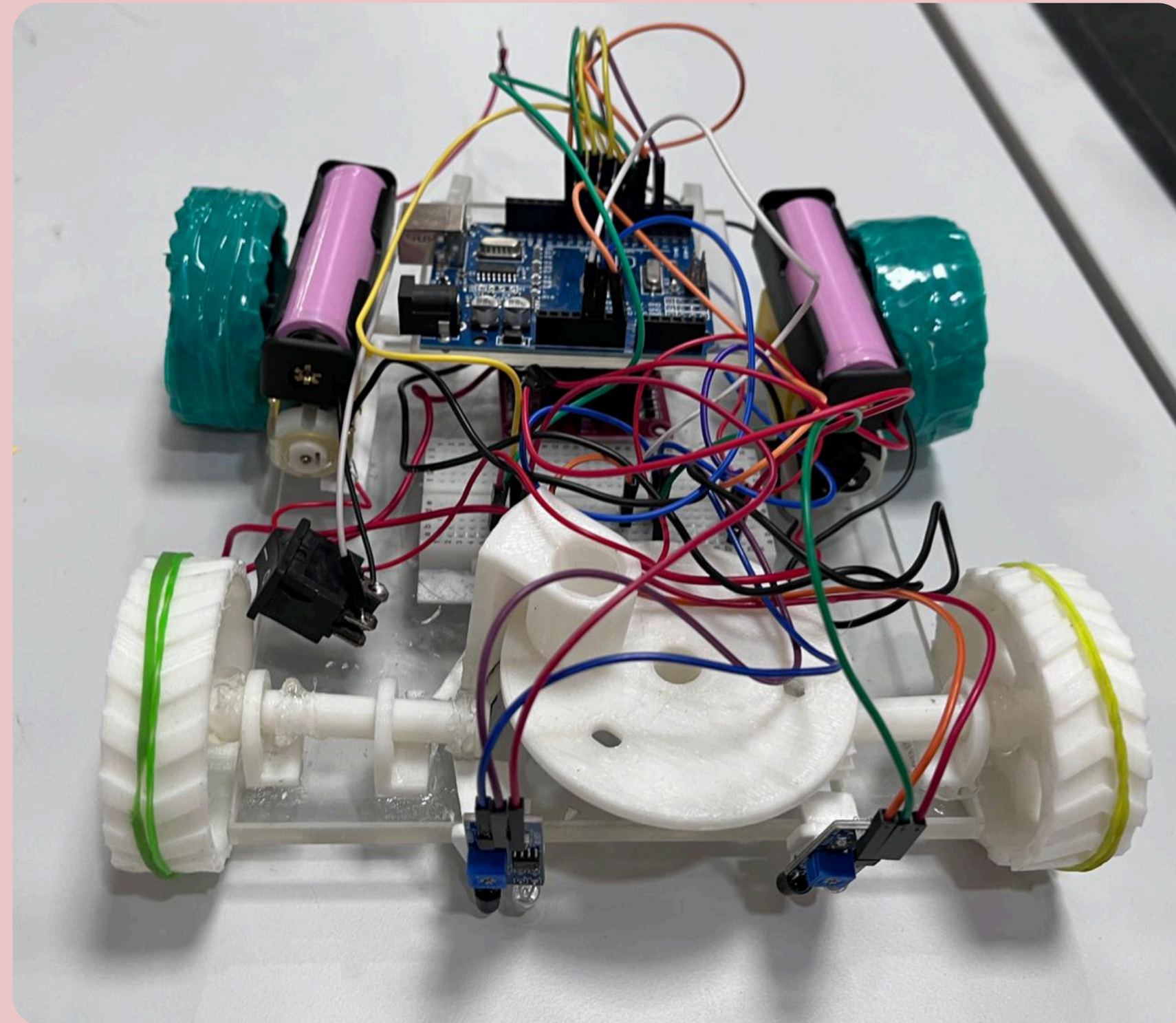
INITIAL TESTING

10



FINAL PRODUCT

11



SUMMARY AND LESSONS LEARNED

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Goal Accomplishment

We successfully developed a line follower robot that accurately tracks and follows lines, showcasing our technical skills and problem solving abilities.

Teamwork and Collaboration

Throughout the project, we cultivated effective teamwork and collaboration, leveraging each team member's strengths to overcome challenges

Continuous Learning

From troubleshooting technical issues to refining our design, we learned the importance of perseverance, adaptability, and continuous improvement.

Proud Achievement

Our line follower robot stands as a testament to our dedication, creativity, and passion for robotics.

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THANK YOU

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